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**ELECTRIC RAILWAYS IN THE OHIO VALLEY
BETWEEN STEUBENVILLE, OHIO, AND
VANPORT, PENNSYLVANIA.***

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WITH DISCUSSION BY MESSRS. F. LAVIS, GEORGE B. PRESTON, J. MARTIN
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There has been constructed during the past two years, along the northerly and westerly banks of the Ohio River, between Vanport, Pa., and Steubenville, Ohio, about 40 miles of double-track, standard-gauge, electric railroad, which affords this busy, thriving, industrial section a high-grade interurban road.

The lines form the connecting links uniting the East Liverpool, Wellsville, Steubenville, and Toronto systems, and, with their connections, at Vanport on the north and Steubenville on the south, make possible through travel by trolley between Pittsburg, Pa., and Wheeling, W. Va., except for a short piece of track, now under construction, in the vicinity of Sewickley.

The configuration of the country through which the road has been built is such as practically to preclude the location and construction of a future competing line.

* Presented at the meeting of January 6th, 1909.

The railways described have been built by three constituent companies:

The Steubenville and East Liverpool Railway and Light Company.—This company owns and operates the properties of the Steubenville Traction and Light Company and the Toronto Electric Light and Power Company. The former of these acquired properties owned and operated the street railway system in Steubenville, and a single-track line to Toronto on the north, a distance of about 10 miles. The new company has reconstructed this single-track road, improving its alignment and grade at various places by locating the tracks on a private right of way; has added a new second track between Steubenville and Toronto, and has built a new double-track road between Toronto and Wellsville, a distance of 7.86 miles. The company also furnishes light and power for commercial purposes in Steubenville and Toronto. At Steubenville, the southerly terminus of the system, connection is made with the Wheeling Electric Railway, which reaches Wheeling, W. Va., a distance of about 21 miles, also with the Steubenville and Brilliant Line, on the west side of the river.

The East Liverpool Traction and Light Company.—This company serves the East Liverpool district, which extends eastward as far as the State line between Ohio and Pennsylvania, and southward to the southerly limits of Wellsville, and includes the street railway system in East Liverpool, a branch line across the river to Chester, W. Va., and a 3-mile spur track to the company's coal mine up Island Run. It also furnishes light and power for commercial purposes in East Liverpool, Chester, and Wellsville. The initial move in improving and increasing the transportation facilities demanded by this territory was made by this company. The road was originally a single-track line, lying wholly in the highways and streets, and had several dangerous steam railroad crossings at grade. In the reconstruction of that portion of the system which forms a part of the through main line, provision was made for a double track, which has been laid partly in new location, and as far as practicable on private right of way, improving the grade and alignment, and eliminating the grade crossings.

The major portion of this property was reconstructed during 1905 and 1906, thus preceding the construction of the stretches of road southward to Toronto and Steubenville, and northward from East Liverpool to Vanport, which has resulted in the continuous line of about 40½ miles of double-track road described herein.

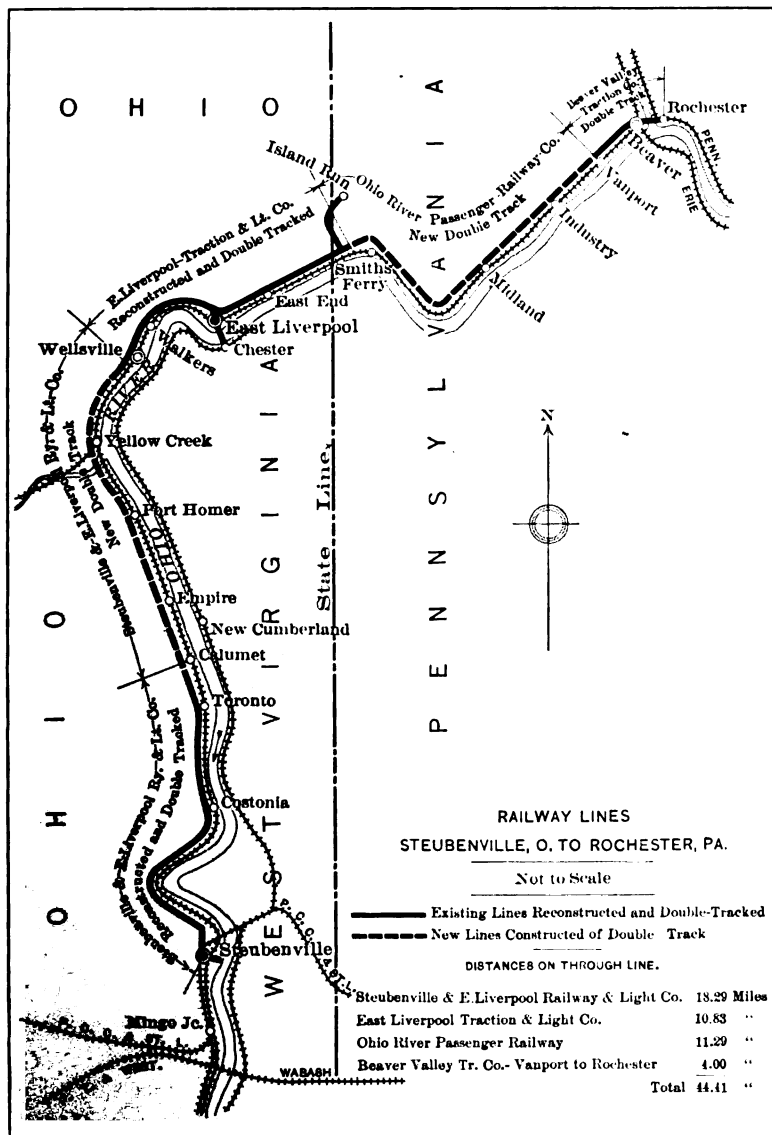


FIG. 1.

The Ohio River Passenger Railway Company.—This company has built the new double-track road between East Liverpool (where connection is made with the lines of the East Liverpool Traction and Light Company) and Vanport, a distance of 11.29 miles. At the latter place connection is made with the lines of the Beaver Valley Traction Company, with which company a traffic agreement has been made for operating cars into Rochester, Pa. Arrangements are being perfected with the Pittsburg and Lake Erie Railroad for a connection at Beaver and for the sale of tickets for through travel to any point on either system.

A tri-party agreement has been executed with the foregoing constituent companies permitting the joint use of the main-line trackage between Steubenville and Rochester, a distance of 44½ miles.

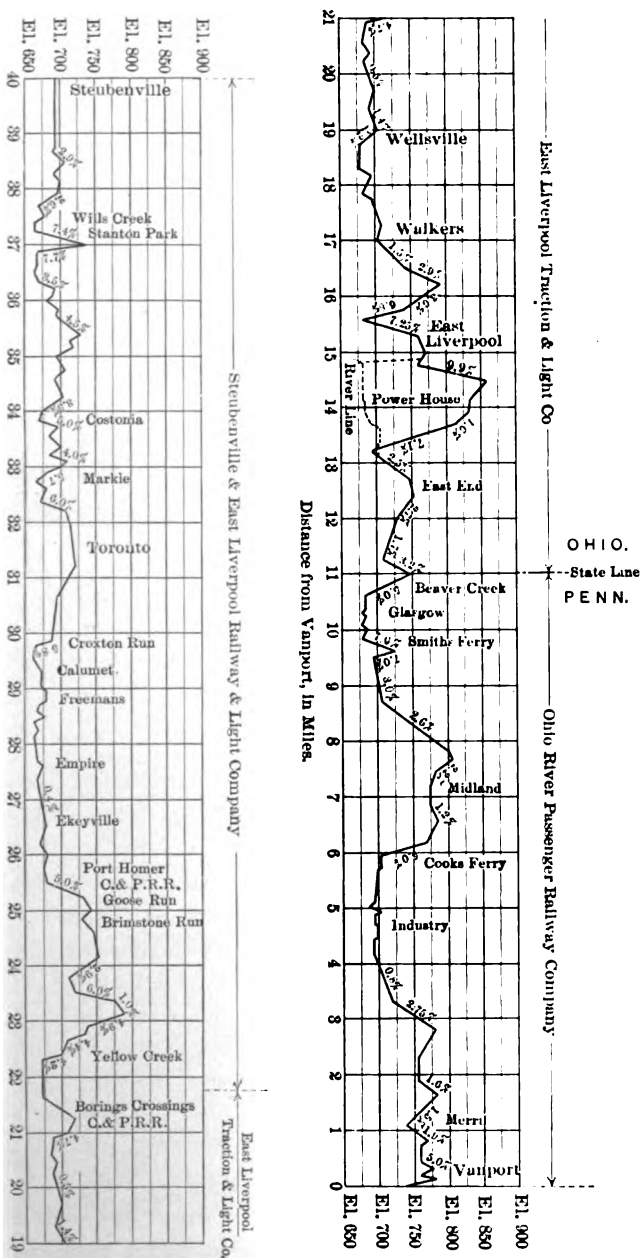
Location.—The portion of the Ohio Valley traversed by these connecting lines is narrow, with high and, in some places, steep and precipitous hillsides and bluffs alternately advancing and receding from the shore of the river.

The location along the bank of the river on a low grade line was practically pre-empted by the Cleveland and Pittsburg Division of the Pennsylvania Lines, and it required considerable engineering study and skill to secure private right of way, and locate a line which would not only give easy curves and grades, but avoid expensive trestles and viaducts.

Railway Construction.—The permanent way consists of a graded roadbed, located partly in the streets and highways, but for the greater portion of the distance on a private right of way, laid with 85-lb. rails, and ballasted in a first-class manner. The alignment and grades are such as are compatible with high-grade urban and interurban roads. All streams and waterways are crossed by substantial bridges and culverts, the former being of steel and the latter of concrete or stone masonry.

Track.—The track is of standard-gauge construction, and in the main is of 85-lb. T-rail, Am. Soc. C. E. section, laid in 60-ft. lengths on 6 by 8-in. by 8-ft. hardwood ties, mostly oak, spaced 24 in. from center to center, and ballasted with gravel or broken stone. The rails are bonded for the return circuit with single 000 protected bonds, except in East Liverpool, where double bonds of the same type are used on each joint. Cross-bonds connect the rails at intervals of 1 000 ft.

PROFILE OF GRADE LINE BETWEEN VANPORT, PA. AND STEUBENVILLE, O.



Distance from Vanport, in Miles.
FIG. 2.

The tracks are generally 9 ft. 8½ in. from center to center; but where bridges or trestles are crossed, provision is made for spacing the tracks 10 or 12 ft. from center to center. Spring frogs of standard pattern are used in all cases outside of city streets. Fig. 3 shows the standard adopted for roadway construction, outside of towns and cities, both in cuts and fills.

Grading.—Outside of cities and towns the grading was practically side-hill work, necessitating some heavy cuts and fills. In many instances the railroad right of way parallels the highway which was originally cut in the sides of these precipitous hills. This roadway was widened, providing room for both the highway and the double-track railway line.

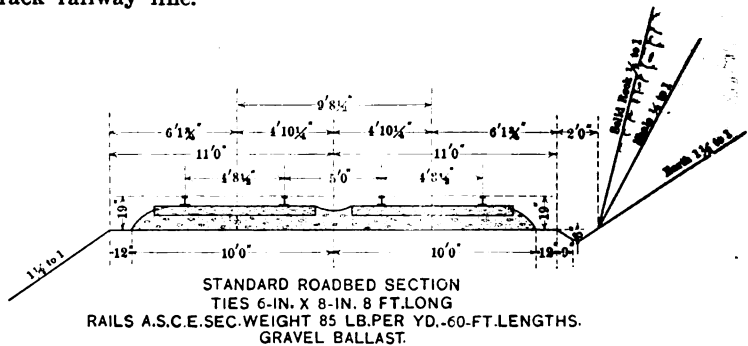


FIG. 8.

The work required the excavation of about 750 000 cu. yd. of material of all classes, a considerable percentage being rock, mostly shale and sandstone.

Beginning at Seventh Street, in Steubenville, the principal work may be described, in sequence toward Vanport, as follows: Grading and paving in highway for track and street widening for 2.76 miles; thence grading and culverts for double track on private right of way across the King and other property for 2.23 miles, this embracing considerable earth and rockwork; thence track grading and highway paving and widening for 1.42 miles; thence earthwork grading and bridging on private right of way through the Toronto Realty Company's property for 0.30 mile to the streets of Toronto; thence grading for track in the streets of that town for 2.2 miles; thence on private right of way and over a double-track, timber trestle at Croxton's Run, 360 ft. in length, for 0.40 mile; thence grading for track, partly in

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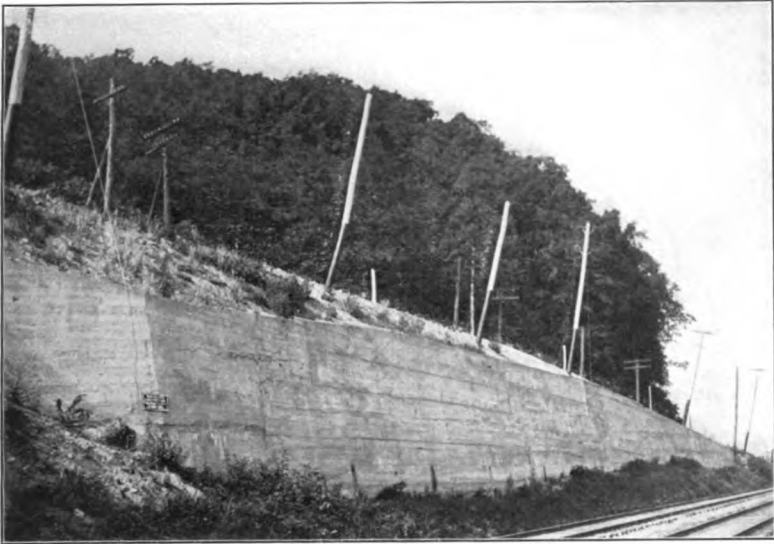


FIG. 1.—RETAINING WALL, COOKS FERRY, OHIO.



FIG. 2.—YELLOW CREEK BRIDGE.

streets, but mostly on private right of way through Calumet, Empire, and Ekeyville for 4.07 miles to the north end of the steel viaduct, at Port Homer, over the Cleveland and Pittsburg Railroad, 450 ft. long; thence side-hill grading on private right of way, including relocation of highways along what is known as The Narrows, to Yellow Creek, for 3.01 miles, this stretch including steel viaducts at Goose Run, 335 ft. long, Brimstone Run, 175 ft. long, various cattle passes, heavy retaining walls, and small bridges; thence by a single-span steel bridge, 174 ft. long, over Yellow Creek; thence under the Cleveland and Pittsburg Railroad through existing bridge openings, and side-hill grading, including the straightening and paving of the highway for 0.78 mile to Borings Crossing, a new bridge being there constructed for the Cleveland and Pittsburg Railroad over the new road and highway, thus removing an existing grade crossing; thence track grading and street paving in Wellsville, for 3.69 miles to the city line of East Liverpool, this stretch including the Wellsville steel viaduct, about 300 ft. long; thence track grading and street paving in East Liverpool for 7.14 miles to the easterly city line, this stretch including the Jethro Run steel viaduct, about 475 ft. long, the Sixth Street steel viaduct, 664 ft. long, Bradys Run culvert, Dry Run bridge, and other structures. For a considerable portion of the latter distance the highways were widened, and some of the new railroad was placed on a private right of way; thence crossing Beaver Creek by a two-span double-track steel bridge, 306 ft. long, and crossing under the Cleveland and Pittsburg Railroad through an existing opening, and in streets for 1.47 miles to a bridge constructed over the Cleveland and Pittsburg Railroad tracks, for both the new railroad and the highway, thus removing a grade crossing; thence track grading on a private right of way, and straightening, widening, and paving the highway for 2.20 miles to Midland; thence track grading and paving for 1.51 miles through this town; thence grading on a private right of way for 1.17 miles to Industry; thence track grading and street paving through this village for 0.37 mile; thence grading on private right of way (mostly side-hill work) for 4.55 miles to the end of the new double track at Vanport. This last stretch embraces the following steel viaducts and other important structures: Four Mile Run viaduct, about 323 ft. long; Barclay Run viaduct, about 275 ft. long, and Vanport trestle, about 700 ft. long.

The side-hill grading was quite difficult, as a great number of slides came down the hillsides, in some instances from considerable distances, causing large expense for their removal. The clayey nature of the material, combined with water from the hills, was the cause of many of these slides. The excavated material consisted of earth, a mixture of earth and rock fragments of varying sizes, and solid rock.

The side-hill work necessitated the construction of a number of retaining walls at different points, the principal ones being at The Narrows, just south of Yellow Creek, and one near Cooks Ferry. The former is about 350 ft. long and has a maximum height of 30 ft.; each contains approximately 1 600 cu. yd. of concrete. The wall at Cooks Ferry is of similar dimensions.

Paving.—The line passes through fifteen communities, and in nearly every instance where a franchise was given for the use of streets or highways unusual requirements for paving were exacted. In some instances the entire width of the street was paved; in others, where the street grade was changed, the embankment slopes in front of residences were paved, as well as the street surface.

As this is a great brick manufacturing center, all paving consisted of vitrified brick on a sand cushion, 11 miles of track being thus paved, the cost averaging approximately \$0.90 per sq. yd.

Bridges.—All important bridges, with the exception of those over Beaver and Yellow Creeks, are deck structures of the viaduct type, weighing about 700 lb. per lin. ft. of double track, and were calculated for a concentrated load of 24 tons on two axles at 10-ft. centers, or a uniformly distributed load on trusses of 1 800 lb. per lin. ft. of each track.

The standard of construction calls for 35-ft. braced towers with battered posts, and 60-ft. spans between the towers, with provision for expansion at each tower. All columns and towers are fixed to the foundations with long anchor-bolts set in the piers, and at the abutments the structure is held in place by short bolts set after the erection of the steel, thus allowing for creeping during erection. The structures, though light, were built for interurban electric passenger service only, and are thoroughly braced and very rigid. The floor systems are stiffened further by carrying the ties clear across the structure. All foundations and abutments are of concrete. Fig. 4 is a typical design for these viaducts.

PLATE XIV.
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FIG. 1.—TRACK VIEW AT INDUSTRY, PA.

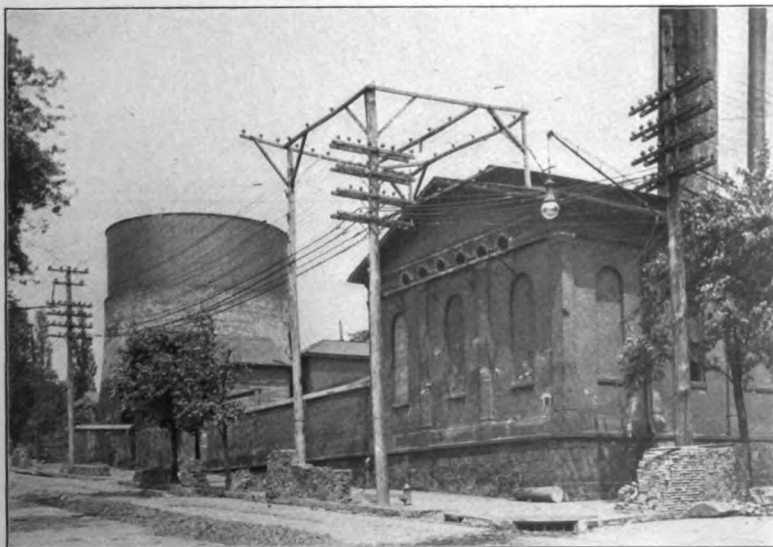


FIG. 2.—WATER-COOLING TOWER, STEUBENVILLE POWER PLANT.

The bridges over Beaver and Yellow Creeks are double-track, through, two-truss structures, having rigid members with riveted connections. The load is distributed evenly over the bearings by using pin-connected shoes. At the expansion end the shoe rests on a nest of turned rollers which are free to move in a planed roller box bolted to the masonry. The structures are secured firmly to the concrete abutments and piers by long anchor-bolts set in the masonry.

The Beaver Creek bridge consists of two double-track, through spans, each 153 ft. long; the Yellow Creek bridge consists of one double-track, through span, 174 ft. long. The steel for these bridges was fabricated and erected by the Fort Pitt Bridge Works.

At Sixth Street, East Liverpool, a steel viaduct was built over what is locally known as Horn Switch. It is 664 ft. 4 in. over all, is 40 ft. above the foundation at its highest point, and weighs about 233 tons. The steel for this bridge was fabricated by the American Bridge Company, and erected by the Cleveland Engineering Company.

The aggregate weight of all the new truss bridges and viaducts approximates 1100 tons.

A number of small I-beam bridges were built, and three of the existing highway structures were strengthened to take care of the loading imposed by the addition of a second track.

Power Requirements.—The power requirements of the original companies forming a part of this property were provided for in four generating stations, located at East Liverpool, Wellsville, Toronto, and Steubenville. These stations supplied current, not only for the street railway operations, but also for municipal lighting and power. To this load factor were added the requirements for the operation of about 60 miles of new track and an increase of 50% in the lighting loads at East Liverpool, Steubenville, and elsewhere.

In the changes, alterations, and additions to the generating, distributing, and transmitting system, the existing houses and equipment were utilized as far as possible.

After a careful study of the conditions, it was decided to consolidate all the generating equipment in the East Liverpool and Steubenville power-houses. The loads formerly carried by the Toronto and Wellsville stations were transferred to these main stations, and the buildings were remodeled and equipped as sub-stations, one new sub-station being built at Industry.

The difficulties of this work were increased by the necessity of keeping the generating stations in continuous operation while the changes and alterations were being effected. The work was accomplished successfully without the interruption of any service that was being rendered.

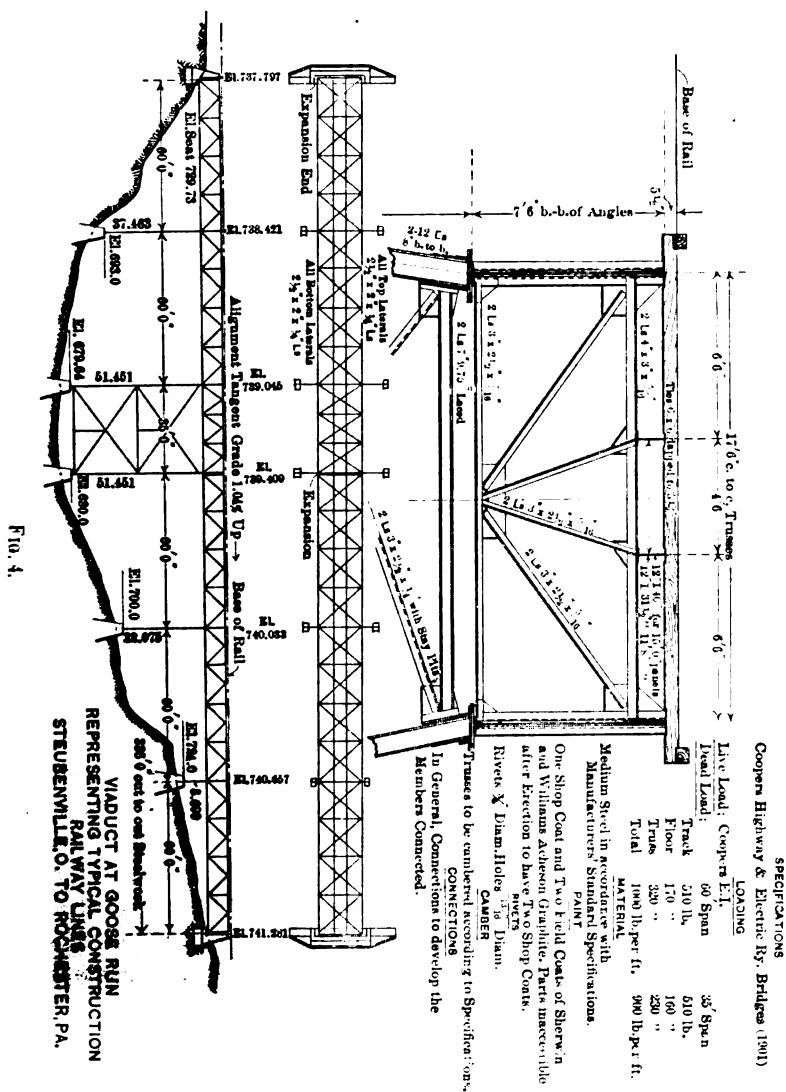
East Liverpool Power-House.—The original building was a brick structure with a wooden floor and a tar and gravel roof on timber supports. The old wooden floor was taken out and replaced with reinforced concrete, giving increased cleanliness and greatly reduced fire risk.

The load on this station was increased by the addition of that formerly carried by the Wellsville station and the new Industry sub-station, as well as by the increased requirements of the railroad, to provide for which the alternating-current generating equipment was increased by the addition of one 1000-kw. turbo-generator and one 500-kw. turbo-generator, and the direct-current equipment by one new 300-kw. rotary converter and the transfer of a 300-kw. engine-type generator from the Steubenville power-house. A new 7-panel switch-board was installed to take care of this additional electrical apparatus.

The turbines as well as the alternating-current, belt-driven generators at this plant produce 60-cycle, 2-phase current at 2200 volts. To reduce this voltage for the operation of the rotary converter, two 175-kw. step-down transformers were installed, and to raise it for transmission to Wellsville and Industry, six 175-kw. step-up transformers were provided. Excitation is taken care of by one steam- and one motor-driven exciter, either of which is of sufficient capacity to furnish excitation for all the alternating-current generators. A Tirrill regulator was installed for voltage regulation.

The boiler plant was enlarged by the addition of a wooden frame structure, sheathed with corrugated iron, in which were installed four 500-h.p. Stirling water-tube boilers. A 2000-h.p., open feed-water heater, and two 12 by 7 by 12-in. boiler feed pumps were also provided.

In order to operate the turbines, a barometric condenser with dry vacuum pump and two motor-driven circulating pumps were installed. Water for condensing purposes is taken from the Ohio River, and, as there is about 40 ft. difference between the high and low stages of the



river, unusual means had to be provided for elevating the water to the condensers. To this end a vertical intake or well of reinforced concrete was built on the bank of the river, extending from below the low-water to above the high-water level. Vertical centrifugal pumps were then installed at the bottom of this well and connected by vertical shafts to two 50-h.p. 2 200-volt induction motors placed above the high-water mark. From the bottom of the head-house a 24-in. intake pipe was extended to the center of the channel of the river, terminating in a timber crib.

The coal supply for this plant is obtained from the company's mine, from which it is carried over a 3-mile branch line and delivered in dump cars at the fire doors of the boilers. The generating capacity of the station has been increased from 1 200 to 3 000 kw.

Steubenville Power-House.—This is a brick structure with a brick floor and a slate roof, supported by steel trusses. A separate building, of similar construction, was enlarged to provide for the installation of two 500-h.p. boilers.

The station load was augmented considerably by the transfer of the lighting load formerly carried at the Toronto station, the increased lighting load for Steubenville, and the additional power required for the operation of the 25 miles of newly constructed railway lines which it serves. To take care of these requirements, the following additional equipment was installed: Two 500-kw. turbo-generators, producing 60-cycle, 2-phase current at 2 200 volts; two 175-kw. and two 100-kw. step-up transformers for raising the voltage to 13 200 volts, 3-phase, for transmission to the Toronto sub-station. The switch-board was enlarged by the addition of four new panels to handle this new equipment. Excitation is taken care of by one steam-driven and one motor-driven exciter, either of which is of sufficient capacity to furnish excitation for all the alternating-current generators. A Tirrill regulator was also installed.

The plant was enlarged further by the addition of two new 500-h.p. Stirling boilers, and the necessary condensing apparatus for the turbines, consisting of a barometric condenser, dry vacuum pump, and engine-driven centrifugal and circulating pumps.

As the only source of water supply for condensing purposes was the city water system, it was necessary to construct a cooling tower and an additional engine-driven centrifugal pump to deliver water

thereto. In erecting this tower, use was made of an old gas tank and well on the property.

The capacity of this station has been increased from 1 800 to 2 500 kw.

Toronto Sub-Station.—The Toronto sub-station building formerly contained the generating apparatus supplying current for the arc- and incandescent-lighting systems of Toronto, and was built of tile blocks, with a brick floor and a slate roof. As now equipped, it contains two 300-kw. rotary converters furnishing direct railway current at 585 volts, and operated from the transmission line through six 110-kw. step-down transformers, and two 110-kw. transformers which deliver 2-phase current at 2 200 volts for the incandescent-lighting circuits, and for the arc-lighting circuits by constant-current transformers and mercury arc rectifiers. The station has a 10-panel switch-board.

Current is received from the Steubenville power-station at 13 200 volts, over two 3-phase transmission lines $9\frac{1}{4}$ miles long.

Wellsville Sub-Station.—The apparatus for this sub-station is installed in a building which was formerly the Wellsville power-station, supplying Wellsville with arc lights for street lighting and incandescent lights for commercial purposes, as well as direct current for street railway operation.

The building is of brick, with a solid concrete floor and a tin roof. Current is received at 6 600 volts from the East Liverpool power-house, over two 3-phase transmission lines each $4\frac{1}{2}$ miles long.

The present sub-station apparatus is designed to take care of the commercial and railway loads, and consists of two 300-kw. rotary converters furnishing current at 585 volts for railway purposes, the current being delivered to the rotaries from the transmission line through six 110-kw. step-down transformers; and four 75-kw. transformers which deliver 2-phase current at 2 200 volts for the incandescent-lighting circuit. The arc lights are taken care of by two constant-current transformers supplied by the 2 200-volt bus. The station also contains a 200-kw. "balancer set," furnishing direct current at 500 volts for a special commercial circuit, supplying shop motors which could not be operated at the railway voltage. The station has a 9-panel switch-board.

Industry Sub-Station.—This sub-station is a new brick structure, with a concrete floor, and a reinforced concrete roof supported by

I-beams. It contains two 300-kw. rotary converters which receive current from the transmission line through six 110-kw. step-down transformers and deliver direct current at 585 volts for railway service only. Current is transmitted to this station at 13 200 volts from the East Liverpool station over a single 3-phase transmission line about 10 miles long. A 5-panel switch-board is installed.

Overhead Construction.—The usual type of span-wire construction used by direct-current interurban roads was adopted. The poles are of chestnut, from 30 to 35 ft. long, with 7-in. tops. They are set 6 ft. in the ground, with head and breast boards, and are given a deflection away from the tracks of 18 in. in 24 ft. The poles are approximately 100 ft. apart, with a minimum clearance of 24 ft. from center to center of poles, measured at the track level. The poles on one side of the road are 5 ft. higher than on the other, and carry, in addition to the trolley, the feeder and transmission lines. The butts of all poles carrying high-tension wires are charred for a distance of about 18 in. above and below the ground line. The tops and gains of all poles were painted with two coats of heavy black asphaltum before erection and before the attachment of the cross-arms. The cross-arms are painted yellow pine, fitted with locust pins, secured to the poles by through galvanized-iron bolts, and braced with galvanized-steel braces.

The trolley lines are mainly of 00 hard-drawn copper wire supported by $\frac{3}{8}$ -in. galvanized stranded span wire to which they are attached by round-top hanger suspensions with 15-in. bronze soldered ears, except on curves, where the ears are carried by gooseneck suspensions with wood strain insulators. They have a clear height above the track of 19 ft. 6 in. outside of city limits and on private right of way, while in the cities this clearance is increased to 21 ft. 6 in. Protection from lightning is furnished by arresters placed on the poles at intervals of about 1 000 ft.

Transmission and Feeder Lines.—The six wires forming the double 3-phase transmission lines between East Liverpool and Wellsville are supported on one 8-ft. cross-arm, the wires of each circuit being 16 in. apart. The double transmission lines between Steubenville and Toronto are carried in delta on two cross-arms, with the apex of the triangle at the bottom, the cross-arms being spaced so that the wires form a 24-in. equilateral triangle. Between East Liverpool and In-

dust, the single 3-phase transmission line is carried on a 7-ft. cross-arm, the wires being spaced 24 in. apart. This construction makes possible, with the least alteration, future provision for a double 3-phase line, similar to the Steubenville and Toronto lines, by simply adding another cross-arm with two insulators.

The wires are carried on brown porcelain, high-tension insulators, 5½ in. in diameter and 4½ in. in height, and are protected from lightning by suitable lightning arresters and choke-coils in each of the power-houses and sub-stations. The lines consist of No. 2 B. & S. hard-drawn copper wire, except between East Liverpool and Wellsville, where No. 4 wire is used. This wire is bare, except through built-up districts, where it is protected by water-proof insulation. At two points on the route the transmission lines are carried on independent pole lines which have been erected for short distances on a private right of way.

In addition to the direct-current railway feeders required for local service, the main line of the road is taken care of by 795 000-cir. mil. aluminum feeders carried on a 2-pin cross-arm below the transmission line, taps to the trolley being provided at intervals of 1 000 ft.

Cars.—New rolling stock has been purchased for the through service between Steubenville and Rochester, and consists of 18 double-truck Pullmans, 44 ft. long, finished in mahogany, with high-backed, red plush seats, having a seating capacity of forty-four. Cars of this type, and for service of this character, are usually fitted for greater carrying capacity, but in this equipment four seats, two on either side, were omitted, the seats being respaced to give more liberal distance between them, following more nearly, in this particular, steam railroad practice.

Each car is equipped with four 60-h.p. motors, mounted on Brill trucks with steel wheels. A smoking compartment, with a seating capacity of sixteen, is provided at one end; all cars are heated and lighted by electricity, and fitted with electric arc headlights, Lecturn signal lights, electric bells, parcel racks, and other conveniences.

The exterior finish is a rich yellow, lettered in aluminum outlined in black. On the letter-board over the windows, the cars bear the words "Ohio Valley Scenic Route."

Organization in the Field for Roadway Construction.—The engineering of those portions of the roadway of the East Liverpool Traction and Light Company's property constructed and reconstructed during 1905 and 1906 was looked after by Westinghouse, Church, Kerr and

Company. The work was executed under various contracts at unit prices by others and by the company's own forces.

The engineering of the subsequent roadway extensions of this company's property, the construction and reconstruction on the Steubenville and East Liverpool Railway and Lighting Company's property, and the work on the Ohio River Passenger Railway Company's property constructed during 1907 and 1908 was looked after and constructed directly by the forces of Westinghouse, Church, Kerr and Company, as was also the power work during 1907 and 1908.

The roadway organization in the field consisted of the engineer in charge, assisted by field parties and superintendents on the different main divisions, with sub-divisions under general foremen, walking bosses, foremen of gangs, etc., these being revised from time to time as the work demanded in its several parts, such as grading, masonry, track-laying, paving, and overhead work. Some unavoidable difficulties in obtaining right of way delayed the work so that the time covered was more than would have been necessary otherwise. The maximum number of men employed at one time was about 1 000. No steam shovels were used or required. Four locomotives, with about 30 dump cars, were utilized more or less continuously in the grading and in the distribution of track material and ballast. There were various concrete mixers, derricks, and pile drivers. The total cost of the roadway work for 1907 and 1908 was about \$1 600 000. The cost of the power work done during 1907 and 1908 aggregated about \$450 000.

Government Work.—Extensive river improvements are being made by the United States Government in the construction of a deep-water channel between Pittsburgh and Cairo and the building of dams and locks which will make navigation possible on the Ohio throughout the year. It is estimated that at the present time, with these improvements still far from completion, more than 10 000 000 tons of freight per annum are moved in boats and barges during the high-water stages, over the 40 miles of river between Beaver and Steubenville.

Population.—Table 1 gives the approximate population of the various cities, towns, and districts through which the road passes, or which, by reason of connections made, may fairly be considered as contributory; also the approximate mileage.

Industries.—The section of the Ohio River between Beaver and Steubenville is rich in resources—coal, natural gas, oil, clay, quarries,

etc.—which make it a natural industrial center. Here are situated some of the large iron, steel, and tin-plate works of the United States Steel Corporation and a number of independent companies; extensive plants for the manufacture of vitrified sewer pipe, fire brick, paving brick, tile and other fire-proofing materials, as well as glass-works and paper mills.

TABLE 1.

District.	Mileage.	Population.
Pittsburg.....	8	700 000
Intermediate.....	17	30 000
Beaver.....	4	46 000
Intermediate.....	11	8 200
East Liverpool.....	11	44 500
Intermediate.....	10	18 900
Steubenville.....	8	41 500
Intermediate.....	19	11 500
Wheeling.....	14	100 000
	102	1 000 600

East Liverpool is the principal seat of the pottery industry in the United States. Three of the twenty-seven potteries located here are said to be the largest in the country; and this industry, which alone employs 12 000 hands, with a weekly payroll of more than \$100 000, produces more semi-porcelain tableware, door knobs, and porcelain electric fixtures, than any other city in the world.

Other important manufacturing interests, related or semi-related to the foregoing, make this 40 miles of river valley "a veritable hive of industry."

Amusement Park.—One of the most beautiful and popular amusement resorts in this section of the country is Rock Spring Park, at Chester, W. Va., just across the river from East Liverpool, and reached by one of the company's branch lines. It constitutes the main pleasure resort in this part of the valley, and no expense has been spared to maintain the high standard of its attractions. It is owned by the East Liverpool Traction and Light Company, and is operated by an amusement company to which it is leased.

Scenic Attraction.—The road passes through an exceptionally picturesque country, and gives promise of excellent possibilities for the development of a large excursion business; it will undoubtedly prove an attraction to pleasure seekers.

For its entire length, the route follows the river, and is located generally on the table lands affording many fine views of the O-hi-yu

or "Beautiful River," which fully justifies the name given to it by the Indian tribes formerly inhabiting its banks.

In the vicinity of Toronto, great apple orchards cover the hills for almost 10 miles, and in the spring the great mass of white and pink bloom against the fresh green of the hillside presents a strikingly beautiful picture. The beginning of the apple industry is traced to the eccentric woodsman, John Chapman, familiarly known as "Johnny Appleseed," who, during the latter part of the eighteenth century, devoted his life to planting apple trees in this vicinity.

Operation.—The various sections of the road were placed in operation as fast as completed, as follows:

Double track between Wellsville and East Liverpool, Sept. 15th, 1906.

Between Calumet and Port Homer, Dec. 14th, 1907.

*Between Steubenville and East Liverpool, Feb. 11th, 1908.

Between East Liverpool and Smith's Ferry, Feb. 29th, 1908.

Between Smith's Ferry and Midland, March 21st, 1908.

Between Steubenville and Calumet, June 30th, 1908.

The entire road between Steubenville and Vanport, July 1st, 1908.

Cars are run on 30-min. schedule on the main through line, with extras and specials as required.

Financiers, Engineers, and Constructors.—The construction of these railway lines has been financed by the Ohio Valley Finance Company, the Hon. W. Caryl Ely, of Buffalo, N. Y., President. The general supervision of the whole project, the securing of right of way, and the purchase of rolling stock was looked after by Mr. Edward McDonnell, of East Liverpool, Ohio, Assistant Treasurer, and Mr. Van Horn Ely, of Steubenville, Ohio, Vice-President of that company.

Westinghouse, Church, Kerr and Company, of New York City, were the engineers of construction and reconstruction of the old properties, and the engineers and constructors of the new properties, the road-way construction being under the supervision and direction of the writer assisted by Edwin J. Beugler, M. Am. Soc. C. E. The execution of the work in the field was under the direction of William V. Polleys, M. Am. Soc. C. E., Resident Engineer. Mr. George B. Preston was the engineer in charge of the mechanical and electrical work in the field and office.

*A part of the distance between Steubenville and Calumet was operated as a single-track road until June 30th, 1908.

DISCUSSION.

F. LAVIS, M. Am. Soc. C. E.—The valley of the Ohio River and Mr. Lavis. the territory tributary to it have witnessed a greater development of long-distance interurban electric railroads than any other part of the United States. Many of these lines run through sleeping, dining and drawing-room cars, built by the Pullman Company, of only slightly lighter construction than the standard equipment of the largest trunk lines of steam railroads. A description, therefore, of one of these lines, completed only about six months ago by one of the most prominent engineering firms identified with work of this kind, is presumably a description of the very latest and best practice in the construction of railroads of this type at the present time.

The general characteristics of these interurban lines are, that they pass through the various cities and towns along the route on the surface of the streets, and as near the business centers as possible, while in the open country between, they are located, for the greater part of the distance, on private right of way, in many instances having numerous highway crossings at grade. This latter feature is not mentioned by the author, and may have been avoided on this road, but the number of these crossings, which are only slightly, if at all, less dangerous than those of steam railroads, is becoming increasingly large, even in States where a great deal of public money has been, and is being, expended in eliminating such features on steam railroad lines.

This road, between Rochester and Steubenville, passes through a territory described by the author as a very "hive of industry," and the population tributary to it, amounting to more than a million people, or practically 10 000 per mile on the whole length between Pittsburg and Wheeling, is composed largely of the class of prosperous mechanics and laborers which furnishes a larger patronage, in proportion to the total population, to a road of this kind than does any other. The most substantial construction, therefore, is apparently warranted.

In order to get some idea of the relative importance of the construction on such a line, it seems desirable to have some basis of comparison with other standard railroad practice. The bridges on this line are designed for a concentrated load of 24 tons on two axles at 10-ft. centers, or 1 800 lb. per lin. ft. of single track. The heaviest equipment of the New York Subways and Elevated Lines has a load of 30 tons on the two axles with the motors, and 11 tons on the other two, or an average of 2 080 lb. per lin. ft. of track, which is only slightly greater than that used on this line. The largest Pullman cars in use on steam railroads are about 80 ft. long, and weigh about 60 tons, this load being carried on two six-wheeled trucks, giving about 10 tons per axle and about 1 500 lb. per lin. ft. of track. A passenger locomotive of the Atlantic type, weighing about 75 tons, which is a fairly

Mr. Davis. large engine for express passenger service on anything but mountain grades, has a load of about 20 tons per axle on the driving wheels, and the load, for engine and tender loaded, will be about 4 000 lb. per lin. ft. of track. A consolidation freight engine, weighing about 100 tons, has a load of about 22 tons on the driving axles, and will average 5 000 lb. per lin. ft. of track. The bridges on this road, therefore, will carry almost any standard railroad equipment except locomotives, the weights of the heavier types of which in ordinary use would be about double that for which these bridges were designed.

Having in view, therefore, the fact that the requirements of roads of this class are not, in many respects, very far removed from standard steam railroad practice on first-class lines, and are in some respects greater than the requirements on many inferior lines, it has seemed to the speaker that it would greatly enhance the value of the paper if the author would give some of the reasons for the very radical departures from steam railroad practice as regards the location of this line.

The author states that "It required considerable engineering study and skill * * * to locate a line * * * which would give easy curves and grades," considering the natural difficulties of topography, and the possibility of securing private right of way. Nothing is stated in the paper in regard to what the author considers to be easy curves for a road of this type, although, as a matter of fact, outside of the cities, curves of from 200 to 300 ft. radius, provided there were not too many of them, and the amount of central angle covered was necessary, would probably have little effect either on cost of operation or speed. In passing through cities, curves of as small a radius as 50 ft. are not infrequent. This difference between roads of this type and trunk-line railroads where a curve of 1 000 ft. radius is considered quite sharp and necessitates slowing down to from 20 to 25 miles an hour, is quite marked, and is due of course to the absence on the electric road of the long rigid wheel base of the steam locomotive.

The question of grades on these electric roads, however, is one about which far less is known, at any rate publicly. Railroads building new lines to be operated by steam, even in a great deal of the, as yet, undeveloped parts of the country, are very reluctant to adopt higher rates than 0.6% unless the topography of the country absolutely demands it, whereas on the profile of the road under consideration, as shown by Fig. 2, one grade of 9.9% is shown for a fairly long stretch, and there are several stretches of 6% to 7%, although most of them seem to be fairly short.

The reason, of course, for the badly broken grade line is the fact that the river bank is occupied by the steam railroad, thus forcing the electric road in many cases high up on the side-hills where these

come close to the river and necessitating its descent nearly to the river level where the hills lie farther back, in order to avoid a long detour to keep up. This characteristic of the country is shown very clearly in the view of the Yellow Creek Bridge. Mr. Lavis.

The ability to ignore, to a very large extent, considerations which affect the determination of the rates of grade, which govern steam railroad practice, is the chief characteristic which differentiates the location of interurban electric railroads, or any electrically operated railroad where single cars or multiple-unit trains are to be used exclusively, from that of those operated by steam or electric locomotives at the head of the train, although few data have as yet been made public as to the actual conditions which govern the determination of the most economical grades to be used on these electric roads.

On a steam railroad, in conjunction with the necessity of low grades, it is also almost as important to arrange the grade line so that the demand on the locomotive will be as nearly even as possible throughout comparatively long stretches. No considerations of this kind apparently influenced the fixing of the grades on this railroad, and here again it would undoubtedly be of considerable interest to know just what considerations governed the judgment of the engineers in adopting the grades they did on this particular road.

As the steam locomotive is most economically operated when the demand on the machine is fairly consistent, so also is this the case with almost any machine or power plant, including, and perhaps especially so, electrical power plants.

In the case of the latter, when power is being manufactured for the operation of a railroad, it may be considered that, where a fairly large number of cars are being operated, the demand will be equalized by the fact that some cars will be going down grade while others are going up, some stopped while others are going at full speed, etc.; but, on a road such as the author describes, where the service is infrequent and the number of cars in use at a time small, the very broken grade line and the high rates of grade would apparently tend to create a somewhat irregular demand, as there would be nothing to prevent a combination frequently occurring where all the cars, or at least a majority of them, would be going up hill at once, and then within a few minutes be all going down hill. In the case of power-houses supplying power only for the operation of the cars on a railroad of this kind, this might be a serious consideration. The power-houses on this line, however, supply power for both manufacturing purposes and electric lighting, for which purposes the load is probably fairly even, and is a large proportion of the total output. The variation in the load from the operation of the cars, therefore, probably does not affect them greatly. The great economy of a uniform load at a power-station is strikingly exemplified by the contract recently made by the Com-

Mr. Lavis. monwealth Edison Company, of Chicago, for supplying power to the street railway companies of that city.* The very low rate of 0.4 cent per kw-hr. was made, provided the demand was uniform, any variation from this uniform demand being penalized according to a certain schedule mutually agreed upon. Of course, a street railway operating in any large city will have high peak loads during the rush hours, and therefore the demand cannot be regular; this instance, however, being so recent, and the contract so important, serves to emphasize the desirability of uniform loads at the power-houses.

It is well to bear this in mind and not lose sight of the fact that the electrical operation of railways has its limitations, although many liberties may be taken with the grades and alignment that would not be permissible on steam railroads.

There is another question which it has seemed to the speaker might naturally occur to some of the readers of the paper, and that is: Before deciding on the construction of these lines at all, was any consideration given to the possible effect on the new line, should the steam railroad, with its truly easy grades and curves, decide to electrify its line and run short trains at frequent intervals, or even equip its line with one of the many types of motor cars now apparently in successful operation on many steam railroad lines, where frequent service for a comparatively small number of passengers at a time is demanded? Probably the steam railroad would be at a disadvantage by its inability to reach the business centers of the various towns, and also to leave every passenger nearly at his own front door. It is quite probable, also, that such a service as that necessary to compete successfully with the electric lines might interfere too much with the through business of the road, both passenger and freight.

In constructing an interurban railroad across country, some years ago, where 60-ft. rails were used, the speaker had some trouble from the excessive expansion and contraction in such comparatively long lengths, until the ballast was filled in as close as possible to the top of the rails and on both sides of them. If the temperature changes are at all large, and the joints are left open sufficiently to prevent the track from kinking in hot weather, there is likely to be excessive pounding of the ends of the rails at the joints, due to the wide spacing. It would be interesting to know whether any such trouble has been found on this road; and, if so, what means were taken to overcome it. The standard cross-section of the track, shown in Fig. 3, seems to show that no ballast was placed above the tops of the ties.

It is quite common practice now, in laying street railway tracks through paved streets, to lay a fairly substantial concrete foundation both under and around the ties, and to such a height over them as will just allow for the type of paving to be used, with the necessary

**Engineering News*, December 10th, 1908.

sand cushion under it. The speaker had occasion to note only a short time ago the installation of this type of construction for a street railway track operated by a large public service corporation, on a double-track line, where the cars were operated on 20 min. headway, with little if any special excursion business, so that the traffic was probably not greater than on the line described by Mr. Francis. No mention of such a type of track construction is made in the paper. If it was not used through the paved streets in the cities, was it on account of the necessity of curtailment expenses, or because it was not considered economical, in view of the expected traffic on the road? The excessive wear and tear on the motors of electric cars due to poor track has led to the very general adoption of the very best type of track construction possible for street railroads almost everywhere.

The width between centers of tracks, adopted on this road, namely, 9 ft. 8½ in., is the general practice on street railway work in cities, though many interurban roads, where high speed is used, have adopted a wider spacing through the open country. On the road recently completed between Baltimore and Washington the tracks were laid 11 ft. on centers, and 80-lb. rails in 33-ft. lengths were used. The fact that the bridges on the Ohio road were designed to allow the tracks to be spaced 12 ft. on centers allows the inference that the engineers had in view the possible necessity of wider spacing of the tracks in the future.

There is another point which the speaker believes is not made quite clear in the paper, and that is the reason for building a double-track road. The only information as to the amount of traffic is the statement that cars are operated on 30 min. headway for ordinary traffic, and that there is a considerable excursion business. On roads with cars operating normally on 30 min. headway, it is generally quite satisfactory to build a single-track road with sufficient sidings to provide for operation on half that headway, the speaker having in mind a road in New England which handles quite successfully a very large summer excursion business on this basis.

The construction of steam railroads is now carried out along lines which have become fairly well standardized, and the type of construction necessary to meet certain conditions is in a general way well known to engineers at all familiar with this branch of the profession. The construction of electric railroads, however, owing to the much greater flexibility of the application of power, is subject to much greater variations, and a most intimate co-relation of the work of the engineers responsible for the location of the line and the electrical engineers is necessary in order to produce the most economical results. The practice in the construction of this road, as described by Mr. Francis, is so radically different from steam railroad practice that the speaker believes that a fuller statement of the reasons which led to the final design, so to speak, of this road, would be most interesting, not

Mr. Lavis. only to the members of this Society, but to all engineers interested in work of this class.

Mr. Preston. GEORGE B. PRESTON, Esq.—As having some bearing on the relation between the average and maximum loads on the power stations, it may be of interest to note that with the heavy holiday load of July 4th, 1908, the 1-hour maximum at the East Liverpool house was 2 000 kw., while the average load during the heaviest part of the day, from 7 to 11 P. M., was 1 800 kw. On an ordinary day the 24-hour average for this station is 850 kw.

When considering the question of peak loads, in connection with this system, it should be borne in mind that these two power-houses are furnishing all the commercial and city lighting, as well as carrying a considerable commercial motor load, for four cities having an aggregate population of 73 000.

Mr. Boucher has asked whether or not the possibility of using the third-rail was considered in connection with this railway system. It was not considered feasible to use the third-rail system because of the necessity of utilizing the local electric railways at Steubenville and East Liverpool which were already installed and equipped with the overhead trolley, and also from the fact that a considerable portion of the roadbed lies in public streets and highways.

Mr. Schreiber. J. MARTIN SCHREIBER, ASSOC. M. AM. SOC. C. E. (by letter).—Although no new engineering features, other than represented in modern practice, are brought out in the paper, it is an intelligent and comprehensive description of a very interesting electric railway.

It is gratifying, at least to those who have had experience in the maintenance of some of the old-time electric railways, to know that the capitalist is rapidly realizing that it even pays to spend some money on the permanent improvement of street railways. Indeed, the very near future will see almost all electric roads constructed with as much skill and forethought as is represented on steam roads. This will be necessary for economical, satisfactory, and profitable operation, and in order to compete with other carrying companies.

The physical difficulties encountered in building the track and roadway of the Ohio Valley lines were out of the ordinary, as the location was not favorable for the work, and the author seems to be justified in the assertion that the configuration of the country through which the road was built is such as practically to preclude the location and construction of a future competing line. No doubt this condition accounts for the variable and at some places severe grades of the track. However, since the road is double tracked, successful and satisfactory operation is facilitated.

The author states that 85-lb. rails, in 60-ft. lengths, and of the Am. Soc. C. E. section, were used throughout the construction and

reconstruction. The writer is of the opinion that rails in 60-ft. lengths Mr. Schreiber. are not altogether desirable on open railway work. It is true that there are fewer joints and more of a continuous track with a longer rail, but there is a disadvantage in handling them, and difficulty in maintaining proper alignment. Unless special expansion joints are used, 60-ft. rails will kink in summer and have open joints in winter. The 33-ft. rail seems to work to the best advantage for open track, and the 60 and 62-ft. lengths for city or paved track.

From one of the photographs it appears that the rail in paved city streets is also of the T-section. The advisability of using the T-rail in paved streets and adopting it as a standard in all street railway work has been strongly advocated by many railroad men. The Way Committee of the American Street and Interurban Engineering Association will make this subject its principal work during 1909, and it is hoped that a definite recommendation will be offered at its next annual convention.

A number of authorities, including the engineers of cities, are co-operating with the street railways in permitting the installation of T-rails in the streets. Several of the up-State New York cities, including Utica and Syracuse, laid T-rails during 1908, and they are reported as very satisfactory; and such cities as Minneapolis, with T-rails and brick paving, claim to have some of the finest street-railway track and roadway.

The great advantage of the T-rail, over any other section, from an operating standpoint, and especially with the increasing wheel loads, is too well known to explain at length, and the question is often only in regard to permission from the authorities for its installation, as they generally object on the ground that the rail does not allow proper paving facilities, and that the paving wears too rapidly along the gauge line of the rail.

It is indeed unfortunate that in the early installation of T-rails in city streets, the paving, especially adjoining the rail, was not properly executed. This placed the T-rail in disrepute for municipal work, so that it is not uncommon for the engineer of a small borough to demand the installation of a Trilby rail, similar to that required in Philadelphia or New York. The writer has often found this to be the case where vehicular traffic was practically nil, and where the paving was macadam. Here the T-rail would not only be cheaper and a great deal more desirable by the railroad company than any other of the common girder sections, but would give better service and more satisfaction to the public at large. This paving about such a rail is as easily maintained as that adjoining the Trilby rail, and with the Trilby section the broken stone is continually getting into the groove and breaking wheel flanges, causing uncomfortable riding, disagreeable noises, and

Mr. Schreiber. frequent derailments, not to mention the trouble of maintaining the line and gauge.

On account of the great improvement made in T-rail track construction, it is the consensus of opinion of railway engineers that it is only a question of time when municipal authorities will co-operate with railway companies in the adoption of the T-rail, except in the very largest cities, where the streets are narrow, and traffic is very heavy. Just how far the T-rail may be used in densely populated cities, and with heavy wagon traffic and narrow streets, where trucks are compelled to follow the tracks, is difficult to anticipate; but, for ordinary city and interurban railways the T-rail should certainly displace the girder section.

The great difficulty of maintaining track and keeping it in line, and with proper joints, with the Trilby and tram sections, are objections which must be overcome; and there are railway managements which, in all probability, will direct their efforts in the future to satisfying the municipal authorities, by building better track, with substantial foundation, proper drainage, and more improved pavements to suit the T-rail, rather than attempt to operate with an unsatisfactory rail that will meet the present paving conditions.

Mr. Boucher. WILLIAM J. BOUCHER, ASSOC. M. AM. SOC. C. E.—This paper is interesting in several particulars, and draws one's attention to the very extended development of cross-country electric roads competing with steam roads in Ohio and Indiana, many of which publish time-tables and schedules, run "limited" trains, and traverse distances from 50 to 100 miles on one division. As these roads are primarily to connect centers of population, they necessarily run through sparsely settled districts, and frequently on private right of way; the query then arises, why are so few roads operated by the "third-rail"?

In any consideration of the cost of installation, it is necessary to specify the kind of work wanted. Single-track, span-wire trolley construction may be had at a variety of costs, but, for similar operating conditions and equivalent current capacities, there is a marked difference in the cost of overhead and third-rail construction. Assuming, then, that first-class construction in both cases is called for; also similar operating and current conditions; a double-pole, single overhead trolley with feed wire; other material and labor necessary for erection may be had at a cost of from \$4 600 to \$5 000 per mile, depending considerably on the nature of the ground in which the poles are placed. An equivalent third-rail system, using an 80-lb. rail on extended ties and insulators, with splice-plates, bonds, and underground cable for road crossings, will cost, erected, about \$3 800 per mile of single track. The third-rail is unprotected. Board protection, similar to that installed in the New York Subway, will cost about \$2 000 per mile. The maintenance charges should be considerably less for the third-rail system over a period of years.

Noticing, then, this difference of \$800 to \$1 200 per mile in favor of the third-rail system, there must be some compelling reasons why the overhead trolley is used so much more frequently. Snow should not cause trouble to a third-rail if the interval between cars is no greater than 30 min., and during a storm any road must be prepared to run all night in order to keep the road open. Sleet, however, may cause trouble, but a scraper has been devised and is in use in New York which seems to overcome the difficulty. Of course, in the case of roads operating through country and towns, the frequent shifting of shoe and trolley would prove a nuisance, especially if a double-throw switch must also be operated, and without the latter the shoe is dangerous to passengers who congregate around the steps and near the trucks while waiting to board the cars. In the case of long-distance roads, however, with expanses of country and few towns, and running alongside a platform at stations, the third-rail would seem to be superior. If the road on its private right of way is properly fenced, and cattle guards are used at crossings, but little trouble will be caused by cattle and horses.

The following roads are now operating with the third-rail: Albany and Hudson; Wilkes-Barre and Hazleton; Lackawanna and Wyoming Valley; Aurora, Elgin and Chicago; Scioto Valley, Ohio.

GEORGE B. FRANCIS, M. AM. SOC. C. E. (by letter).—Referring to the discussion by Mr. Lavis: There were a few grade crossings created on the private right of way portion of the railways, but these could not be avoided with reasonable economy, and their creation was not prohibited by law or the public authorities.

The railways described are of the mixed type, comprising electric city street railway and interurban railway, for passenger service only. These conditions oblige and warrant radical departures in lines and grades from steam railroad practice, even on the private right of way portions. These portions, however, were located so as to conform substantially to steam railroad practice, taking into consideration character of power, equipment, and service. Speeds on lines of this character, operated with the type of wheel flanges required in usual street railway service, are necessarily less than would be the case with master car builder's steam railroad wheel flanges.

The steep grades indicated on the profile occur in places where the line is in public streets, and are governed thereby. Private right of way location was out of the question in such places, for many reasons, mainly because it was desirable to occupy certain main streets in best serving the public and increasing the revenue.

Consideration was given to the possibility of the parallel steam railroad improving its passenger facilities by electrification or otherwise, but, for various reasons, the probability of their so doing was thought to be too remote to govern the construction or non-construction of the new line; the main reason being the interference of local

Mr. Francis. passenger service with the requirements for freight service, in this particular locality, on the steam railroad lines.

No trouble has been experienced from expansion of the 60-ft. rail length adopted on this road. Such troubles, however, have occurred in situations of peculiar character elsewhere.

No concrete foundations were used for either the track or street paving involved. The local street vehicular traffic did not seem to require the placing of such foundations under the paving, the brick paving generally used in this locality, laid on a sand or gravel ballast, appearing to retain a good surface after many years' use.

The road was double tracked for several reasons, viz., safety in operation, adherence to schedule, adaptability to variable schedules (such as are demanded by holidays or changing conditions of service), and because such was the strong desire of the management controlling the properties.

By referring to *Engineering News* of December 10th, 1908, it will be seen that Mr. Lavis' statement: "the contract recently made by the Commonwealth Edison Company, of Chicago, for supplying power to the street railway companies of that city," providing for a rate of 0.4 cent per kw-hr., conveys a wrong impression (probably unintentional on the part of Mr. Lavis), and that an additional payment of \$15 per annum per kw. of maximum demand is provided for in that contract; and this brings the cost to about 0.74 cent per kw-hr. The steady load making this figure possible is not the railway load, but the general lighting load of the Edison Company, which equalizes the railway peaks.

Mr. Preston, in his discussion, appears to have covered the point about the third-rail raised in Mr. Boucher's discussion.